AMAZON VS. FLIPKART: COMPARATIVE MARKET ANALYSIS

An integrated study combining Microeconomics (CO1), Business Environment Dynamics (CO2), and Python Data Analytics (CO3).

INTRODUCTION TO THE TOPIC: OVERVIEW OF INDIAN E-COMMERCE

- **Overview of the Indian E-Commerce Industry:**
- **Unlike the global e-commerce market, which is dominated by Amazon, eBay, and Alibaba, the Indian market is a duopoly of Amazon India (owned by the American company Amazon) and Flipkart (owned by Walmart).**
- **These two companies control more than 70-80% of India's online retail marketplace, vying for the market share in electronics, clothing, groceries, and big-screen TVs, offering steep discounts during the year-end holiday sales events known as Big Billion Days (for Amazon) and Great Indian Festival (for Flipkart).**
- **In essence, the competition between Flipkart and Amazon in India is intense, with both vying for the number one position on multiple fronts**
- **India's E-Commerce Industry Scope:**
- **The study will examine the e-commerce market under a magnifying glass, comparing the companies' market positions, pricing policies, macroeconomic factors, and potential challenges with evidence-based calculations.**

MOTIVATION FOR CARRYING OUT THE PROJECT

- **Massive Market Growth: India's e-commerce market is set to grow to $350 Billion by 2030, making it one of the most important economic drivers in the country.**
- **Duopolistic Competition: The research will shed light on how the two giants that have different development paths (Global and Domestic Born) fight for the largest share of the market.**
- **Price Sensitivity: The paper will discover how customers are influenced and how their behavior is shaped when it comes to sales, flash discounts, and other loyalty programs.**
- **Integration of Data Science: The study will integrate current technologies, advanced Python programming, and specific data science models to address real-life cases and solve existing issues in the field instead of focusing on abstract theories.**

OBJEVTIVES OF STUDY

- Analyze Economic Drivers: Study pricing strategies, demand elasticity of prices, and market forces between Amazon and Flipkart.
- Evaluate External Business Environment: Consider the impact of Government policies (FDI limits, Consumer Protection Rules), and regulatory frameworks, and analyze the PESTLE factors on operations.
- Perform Data Mining with Python: Use Python to gather, clean, and visualize electronic commerce trends (e.g., prices, ratings, discount percentages, and reviews) for both companies, and analyze data trends.
- Conclude Strategic Insights: Synthesize the results to conclude insights about the market situation and future outlook for both companies.

ECONOMICS (CO1): MARKET STRUCTURES & PRICING ELASTICITY

- Topic Focus Duopoly dynamics, Price Elasticity of Demand, Penetration Pricing, Game Theory tactics during festival sales
- Why Selected E-commerce customer behavior in India is highly price-sensitive. Economic strategies can explain how competing players use dynamic algorithms to capture consumer surplus.

BUSINESS ENVIRONMENT (CO2): PESTLE AND REGULATORY ISSUES

- FDI regulations in the context of multi-brand retail, FEMA regulations, anti-trust issues (CCI), local and international supply chains

Reason for Selection: The regulatory environment in India and its impact on business operations e.g. restrictions on a platform to hold inventory of exclusive sellers

PYTHON DATA ANALYTICS: CO3 – DATA MANIPULATION & VISUALIZATION ESSAY

- Python Data Analytics (CO3): Data Manipulation & Visualization Description: The study of data collection (pandas, BeautifulSoup), data cleaning/handling missing values, and visual examination of data (matplotlib, seaborn).
- Why I Chose This Course: Modern businesses need to back their choices with empirical data. Python allows one to quantitatively compare prices of similar products, perform a sentiment analysis of reviews, and much more.

PROJECT APPROACH & METHODLOGY

- Project Approach & Methodology
- Primary/Secondary sources refer to web scraped public product listings (electronics, fashion) and Kaggle datasets with pricing, ratings, and reviews for Amazon and Flipkart.
- Data Cleaning & Preprocessing:
- The cleaning process involved treating missing values, converting currency strings to float data types, and removing duplicates with Pandas.
- Data Analysis & Visualization:
- Inferential analysis was performed in Python to visualize average ratings, discounts with box plots, and relationships between price and rating using Matplotlib and Seaborn.
- Findings & Syntheses:
- The results were synthesized with Python statistical models and economic theory to interpret trends and provide recommendations within macroeconomic and regulatory contexts.
- Final Report Preparation:
- The presentation was compiled into a slideshow outlining key findings and strategic implications for the market analysis.